ITAI 1378 FINAL PROJECT

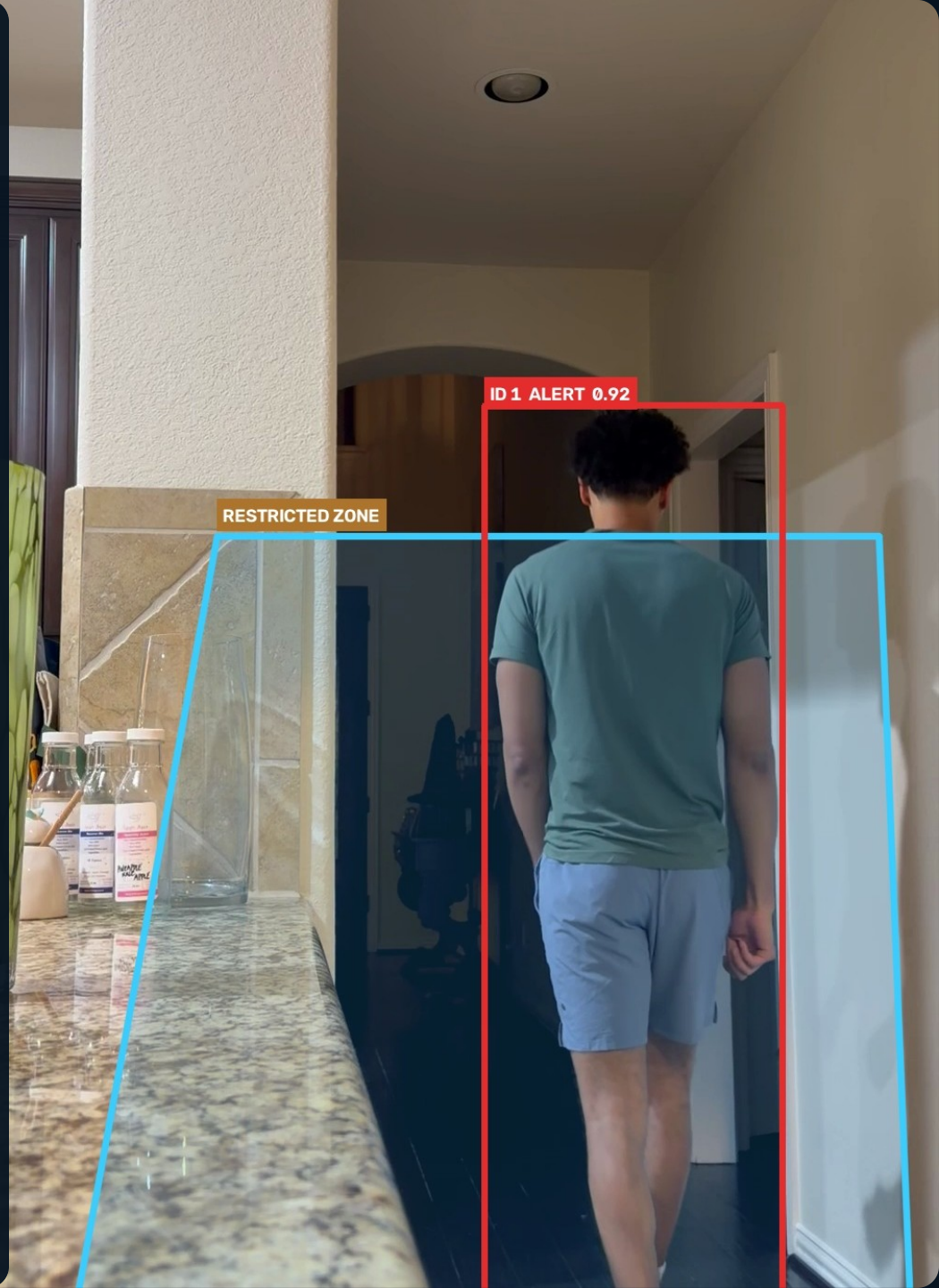
AI-Powered Restricted-Area Monitoring

Working proof of concept using YOLO11n, ByteTrack, and a polygon entry rule

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Tier 2 Advanced

3 videos | 1,250 frames | 4 confirmed events



The problem

NEED

A camera can record a hallway, but someone still has to notice when a person enters the restricted area.

USER

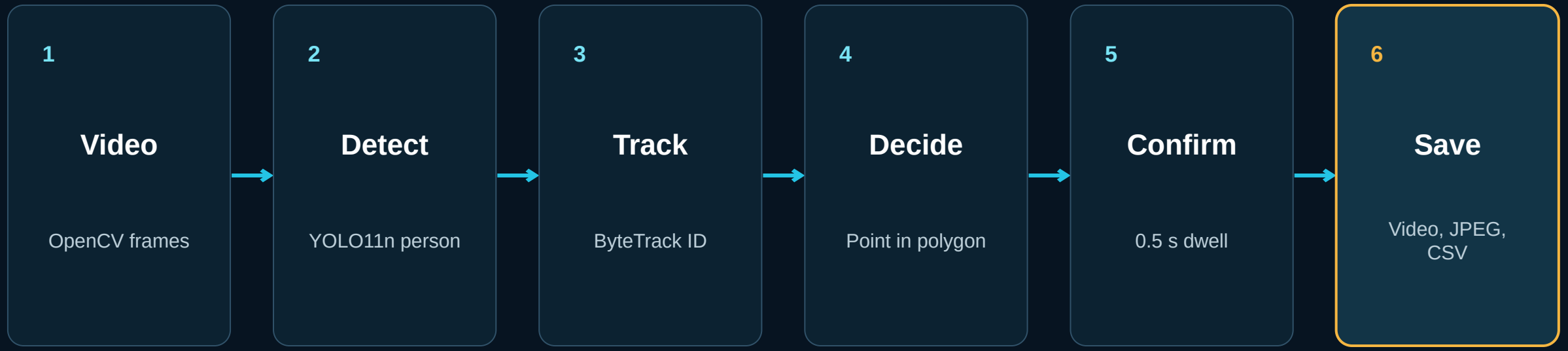
A small office, lab, or school staff member who wants a local event record without watching the video continuously.



Fixed phone camera view used for all three tests

Design boundary: detect entry only. No face recognition or identity decision.

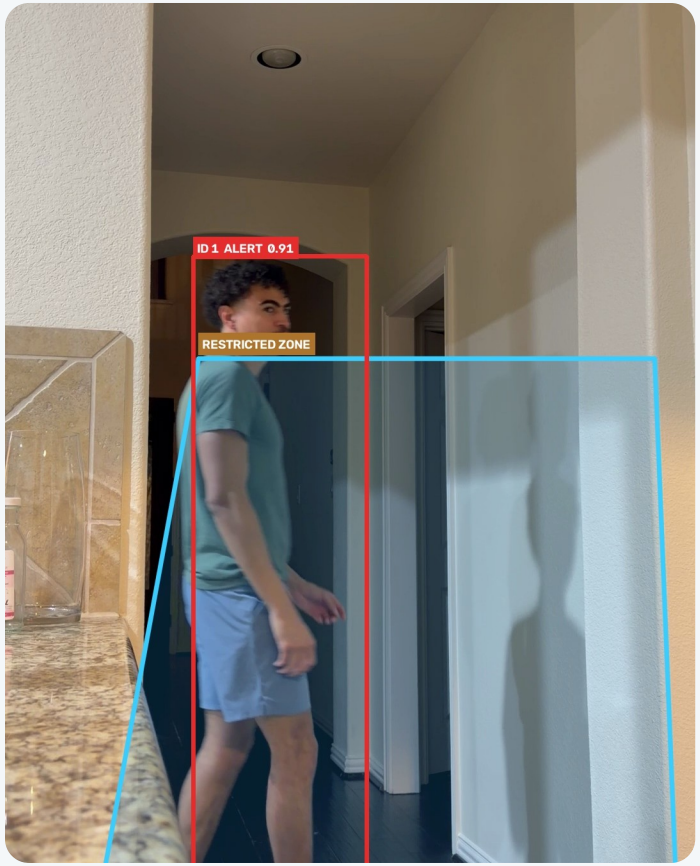
What I built



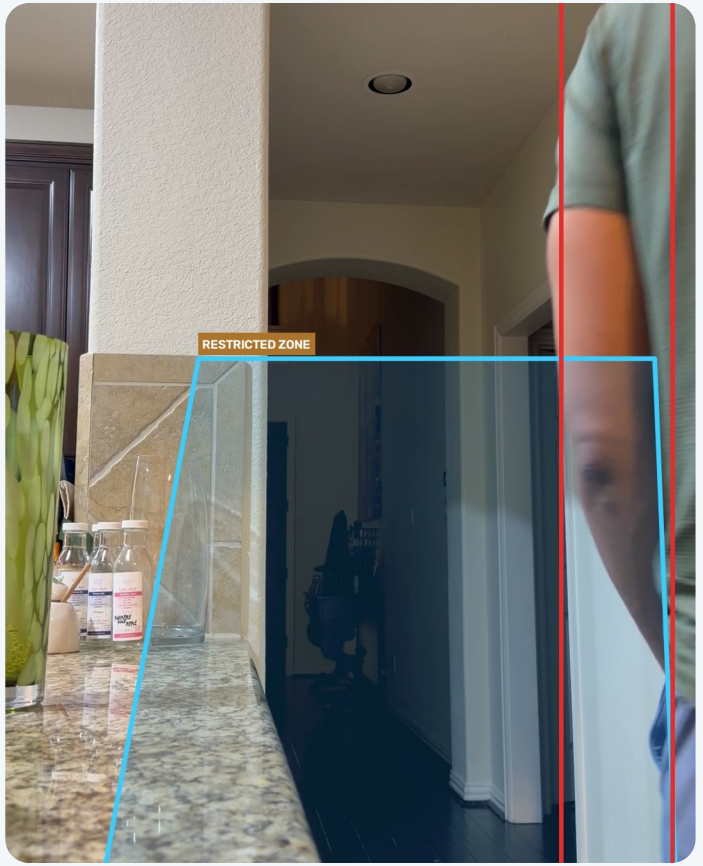
One event per tracked entry. The event resets only after the person leaves the zone.

Actual final pipeline. Every component ran on all three uploaded videos.

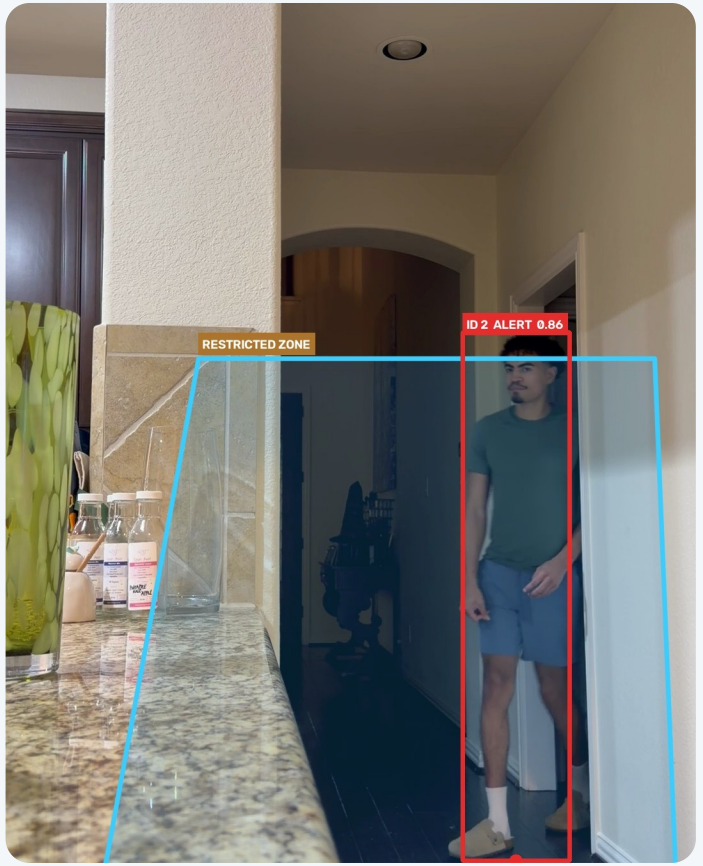
Demo evidence from the real run



Clear entry



Close-camera case



Exit and re-entry

Play results/demo_reel_no_audio.mp4 or the annotated videos during the presentation

How the system works

MODEL AND FRAMEWORK

Ultralytics YOLO11n
Pretrained COCO person class
OpenCV video input and output
ByteTrack multi-object tracking

DECISION RULE

Confidence threshold: 0.35
Image size: 640 px
Floor proxy: box bottom center
Dwell threshold: 0.5 seconds

Core logic

```
if person_detected:
    track_id = ByteTrack(box)
    foot = bottom_center(box)

    if foot_inside_zone:
        inside_frames += 1

    if inside_frames >= fps * 0.5:
        write_alert_once()
```

No training or fine-tuning was required for this proof of concept.

The data I used

3	41.7 s	1,250	4
self-recorded clips	total video	video frames	manual event labels

CLIP-LEVEL GROUND TRUTH

IMG_5462	16.2 s	Clear entry	**1**
IMG_5463	7.3 s	Close, partly cropped	**1**
IMG_5464	18.1 s	Exit and re-entry	**2**

No public dataset, custom box labels, or model training were used in the final build.

Raw phone videos stay local and are excluded from the public repository.

Results compared with the Blueprint

Metric	Blueprint target	Measured result	Status
Person mAP50	> 0.70	Not measured	**No box labels**
Event recall	> 90%	100% (4/4)	**Met**
False alerts	< 1 per hour	0 in 41.7 s	**Too little video**
Event latency	< 500 ms	Worst p95: 18.9 ms	**Met**

The system met event recall and latency targets on this small test. It did not validate mAP or a one-hour false-alert rate.

Weighted mean end-to-end time: 19.9 ms per frame, including startup overhead

What changed from the Blueprint

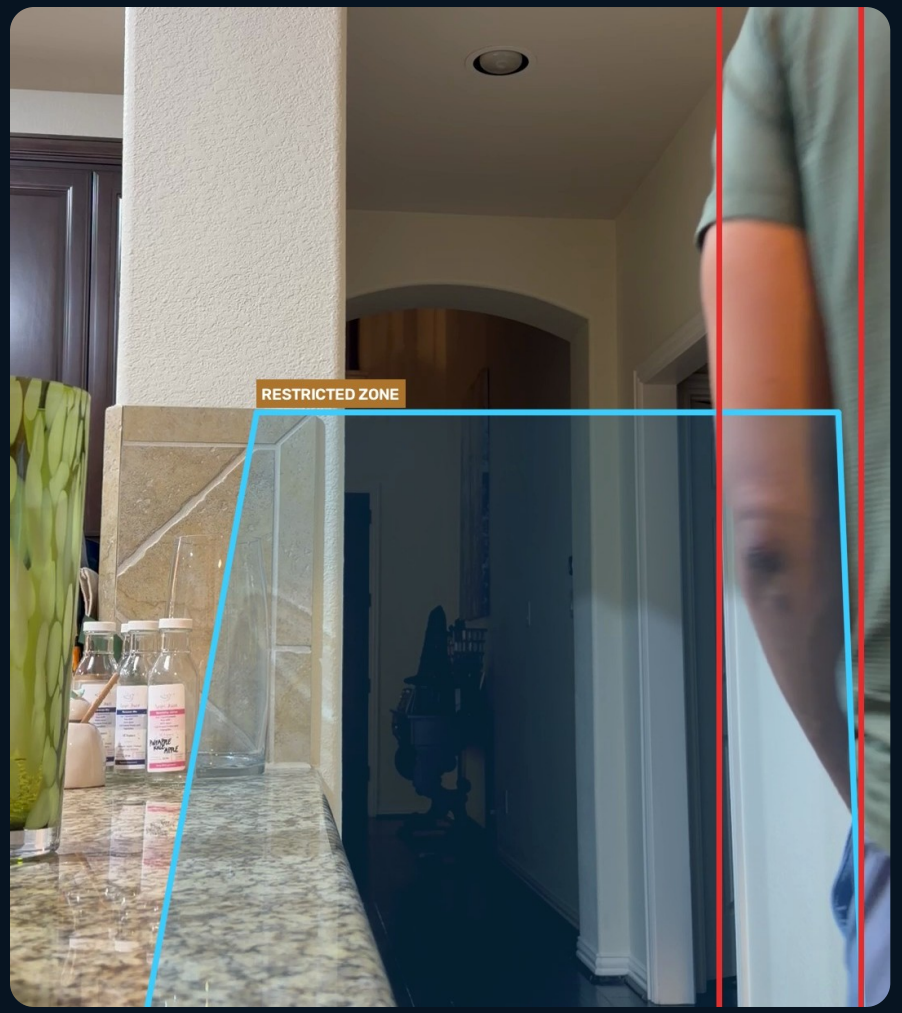
	BLUEPRINT PLAN		FINAL BUILD
Data	Planned public images and staged labels	→	**Used three self-recorded videos**
Training	Fine-tune if baseline struggled	→	**Kept pretrained YOLO11n**
Evaluation	Planned mAP plus event metrics	→	**Measured event behavior and latency**
Tier	Called it a prototype tier	→	**Clarified Tier 2 from course guide**
Deployment	Possible dashboard and notifications	→	**Kept local JPEG, CSV, and video outputs**

Reason: the pretrained baseline already supported an end-to-end test, so I focused on application logic and honest measurement.

Challenges and fixes

	PROBLEM	FIX
Portrait phone video	The model accepted 1080 × 1920 frames, but full-resolution output was large.	**Infer at 640 px and write a 720 × 1280 annotated copy.**
Intermittent close detection	The person was partly outside the image in IMG_5463.	**Use tracking plus a 0.5 second dwell window instead of one-frame logic.**
Exit and re-entry	A person can leave and return during one clip.	**Reset the alert only after the tracked person leaves the polygon.**
Metric mismatch	The Blueprint promised mAP without final box labels.	**Mark mAP not measured and report event-level ground truth instead.**

Struggle case: person close to the frame edge



WHAT HAPPENED

The person was visible in only 27 of 219 frames

Detection coverage: 12.3%

Why it struggled

Most of the body was outside the image, so the detector had less visual evidence and the track appeared intermittently.

Why the event still worked

The remaining detections lasted long enough to satisfy the 0.5 second dwell rule.

Risk: a faster close pass could be missed. Next test: more edge-of-frame negative and positive clips.

Lessons and next steps

WHAT I LEARNED

Application logic matters as much as the detector.

Event metrics require manual ground truth.

Short tests can show function but cannot prove long-term reliability.

NEXT BUILD

Record at least one hour of negative video, test more camera angles, and add an interactive zone-calibration tool.

BEFORE SUBMISSION

1. Upload this repository to GitHub
2. Record a 3 to 5 minute narrated demo
3. Add both public links below

GitHub: ADD LINK
Demo: ADD LINK

Final presentation file: FP_Walters_Kameron_ITAI1378.pdf